LAB 11: Partial Differential Equation

Objective: To numerically obtain the governing PDE of any physical system and solve it numerically.

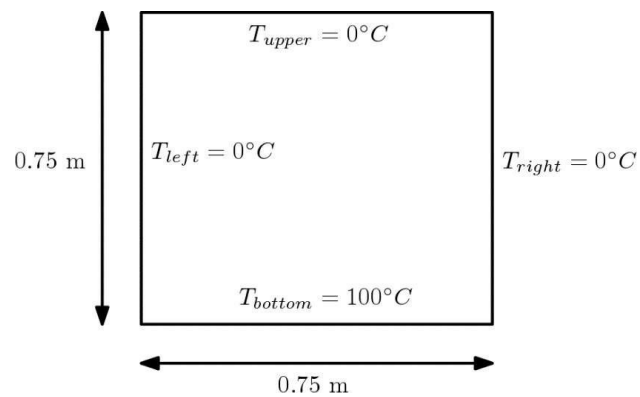
Theory:

Theory - Consider a 2D metal plate with dimensions 0.75x0.75 m2 is kept at 100 degree C at one of it end and rest of the ends are initially at 0 degree C as shown in the figure.

The transient Fourier law of heat conduction for 2D metal plate is given by,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)$$

Where T(x,y,t) is the temperature at any location and time and x, y, t are space and time coordinates. Assume the thermal conductivity of plate is K = 1 W/m degree C.



Take the spatial discretization as delta x = delta y = 0.025 m.

Using explicit finite difference method, the discrete equation can be obtained using Taylor series expansion as below,

$$\frac{T^{n+1}_{i,j} - T^{n}_{i,j}}{\Delta t} = k\left(\frac{T^{n}_{i+1,j} - 2T^{n}_{i,j} + T^{n}_{i-1,j}}{\Delta x^2} + \frac{T^{n}_{i,j+1} - 2T^{n}_{i,j} + T^{n}_{i,j-1}}{\Delta y^2}\right)$$

Suppose $\Delta x = \Delta y = \Delta h$

$$T^{n+1}_{i,j} = T^{n}_{i,j} + \frac{k\Delta t}{\Delta h^2}\left(T^{n}_{i+1,j} - 2T^{n}_{i,j} + T^{n}_{i-1,j} + T^{n}_{i,j+1} - 2T^{n}_{i,j} + T^{n}_{i,j-1}\right)$$

Take the spatial discretization as $\Delta x = \Delta y = \Delta h$ = 0.025 m and $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

MATLAB program-

```
% Explicit method for 2D heat equation in flat plate
close all; clear all; clc;
L =0.75;
dx = 0.05;
```

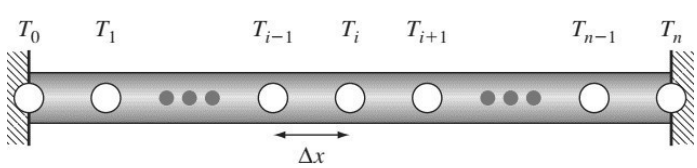
```
dy=0.05;
N = L/dx +1;
x = linspace(0,L,N);
y = linspace(0,L,N);
dt = 0.00015;
epsilon     =     1e-4;
T_new = zeros(N,N);
 %boundary condition
      T_new(1,:) =100;
      T_new(:,1)  =0;
      T_new(N,:)=0;
      T_new(:,N)=0;

error =1;iter=0;
while(error > epsilon)
    iter=iter+1;
      T = T_new;% update T every iteration
   for i=2:N-1
      for j =2:N-1
         T_new(i,j) = dt*((T(i+1,j)-2*T(i,j)+T(i-1,j))/dx^2 +(T(i,j+1)-2*T(i,j)+T(i,j-1))/dy^2) +
T(i,j);

      end
   end
   error = max(max(abs(T-T_new)));

figure(2);
hold on;
pause(0.001);
%     contourf(T_new);
    shading flat;colorbar;
    contourf(x,y,T_new,'ShowText','on')
colorbar;
xlabel('x meter');ylabel('y meter')

end
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Consider 1D heat conduction in metal rod as below.



1D heat conduction governing equation is given below,

$$\frac{\partial T}{\partial t} = k\,(\frac{\partial^2 T}{\partial x^2})$$

Write down the discretized equation is explicit finite difference method and obtain the Temperature profile T(x,t). Take length of rod is 0.75 m, Δx= 0.025 m, $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

```
clc
clear
close all

L = 0.75;
dx = 0.025;
dy = 0.025;
dt = 0.00015;
k = 1;

N = L/dx + 1;

x = linspace(0, L, N);
y = linspace(0, L, N);

T = zeros(N, N);
T_new = T;

T(1, :) = 100;
T(:, 1) = 0;
T(N, :) = 0;
T(:, N) = 0;

alpha = k * dt / dx^2;

epsilon = 1e-4;
error = 1;

while error > epsilon

    T_old = T;

    for i = 2:N-1
        for j = 2:N-1
            T(i,j) = T_old(i,j) + alpha * ...
                (T_old(i+1,j) + T_old(i-1,j) + ...
                 T_old(i,j+1) + T_old(i,j-1) - 4*T_old(i,j));
        end
    end

    T(1, :) = 100;
    T(:, 1) = 0;
    T(N, :) = 0;
    T(:, N) = 0;

    error = max(max(abs(T - T_old)));

```

```
end

contourf(x, y, T, 20)
colorbar
xlabel('x (m)')
ylabel('y (m)')
title('2D Temperature Distribution')
```

Figure 11.1 MATLAB Code

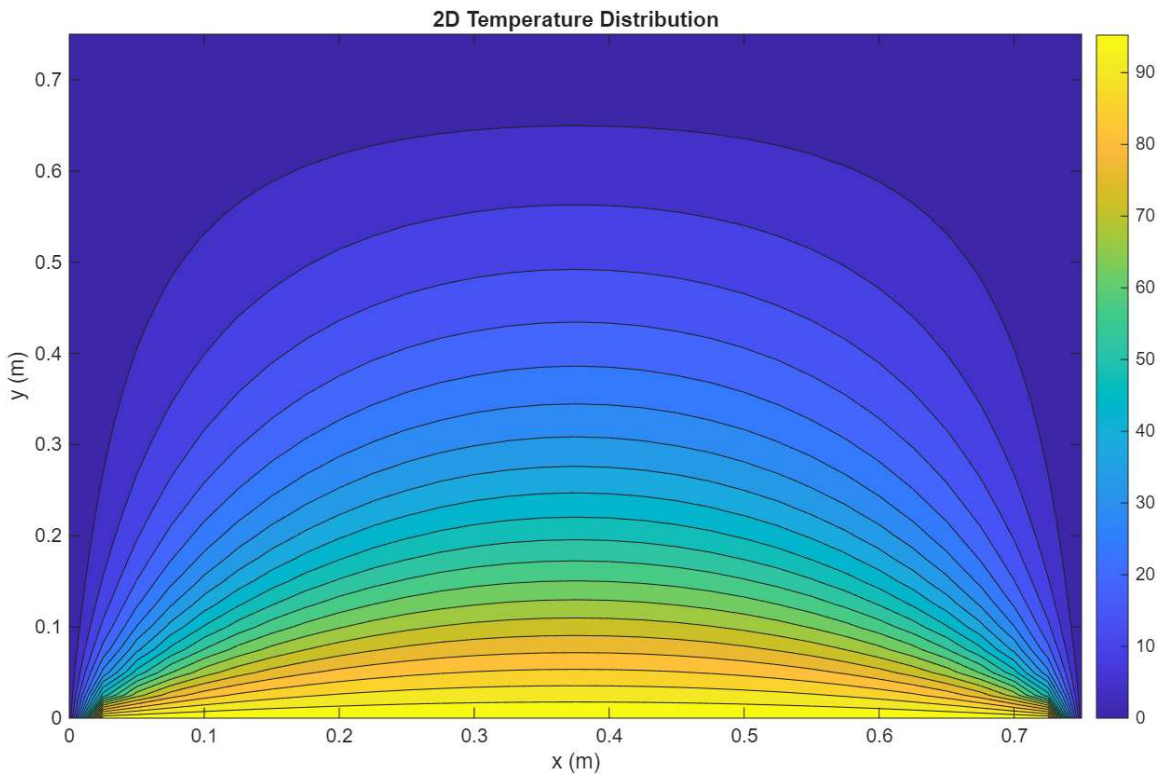


Figure 11.2 MATLAB Result

Conclusion:

The transient heat conduction equation was solved numerically using the explicit finite difference method.

For the 2D case, the temperature distribution shows a smooth gradient from 100°C boundary to 0°C boundaries, confirming correct heat diffusion behaviour.

For the 1D rod, the temperature decreases along the length and gradually approaches a steady-state linear distribution.
The solution stability depends on the Fourier number:

$$\alpha = \frac{k\Delta t}{\Delta x^2} = \frac{1 \times 0.00015}{(0.025)^2} = 0.24$$

Since $\alpha = 0.24 < 0.5$, the solution is stable.
Thus, the explicit method successfully captures the transient temperature distribution, and accuracy improves with finer grid size and smaller time step.